NS Phonon Effects for GW190425

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PAPER_965: NS Phonon Effects for GW190425

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Source: et_phonon_resonance.py §8 (NSPhononGW190425)
Calculator: NSPhononGW190425Calc (CP4 #549)
CVW: v2.0.0 compliant

Abstract

We derive neutron star phonon effects for GW190425 (mass-gap BNS merger). The UQFF phonon-corrected strain $h_\text{UQFF}(t) = h_\text{GR}(t) \cdot 0.5297 \cdot \exp([SSq] \cdot t/26)$ produces a 47% strain reduction. The phonon-corrected tidal deformability $\Lambda_\text{UQFF} = \Lambda_\text{GR}(1 + \delta\Lambda)$ matches LIGO constraints. Mass-gap probability: P(NS) = 49%, P(BH) = 51%.

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1. Phonon-Corrected Strain

$$h_\text{UQFF}(t) = h_\text{GR}(t) \cdot 0.5297 \cdot \exp\!\left(\frac{[SSq] \cdot t}{26}\right)$$

2. Wavelength Correction

$$\lambda_\text{UQFF} = \lambda_\text{GR} \cdot \left(1 - \frac{F_{UBi}}{F_U} \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)\right)$$

3. Tidal Deformability

$$\Lambda_\text{UQFF} = \Lambda_\text{GR} \cdot (1 + \delta\Lambda_\text{phonon})$$

4. Mass-Gap Classification

At $\Gamma = 0.1$ THz: P(NS) $\approx$ 49%, P(BH) $\approx$ 51%.

References

1. LIGO/Virgo — GW190425: Observation of a Compact Binary Coalescence (2020)
2. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
3. PAPER_967 — NS Phonon Tidal Deformability
4. PAPER_964 — 3D MUGE Magnetar Sim
5. PAPER_949 — BCS Gap Equation

Cross-Links

Related Paper	Relationship
PAPER_967	Tidal deformability uses this phonon strain
PAPER_964	Magnetar SCm model feeds $\Delta(r)$
PAPER_955	Phonon Q-factor at ω_SCm
PAPER_949	BCS gap in h_UQFF correction

<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> *Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022*
> *(GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for*
> *GW170817/GW190425 upgraded analyses.*

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47\,\frac{\Phi(\Gamma)}{S_{26}^{(3)}}\right)$$

where:
- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \times 1.25\;\text{THz}$ is the SCm phonon resonance frequency
- $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26}\left[1 + [\text{SSq}]\cdot e^{-\kappa\,i\,n/26}\right]$ is the third-order Ramanujan summation
- Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5\,M_\odot$.

Calibration (canonical): $\kappa = 5 \times 10^{-4}\;\text{day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> *Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),*
> *PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow*
> *Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044*
> *(Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).*

The SCm phonon field introduces a buoyancy force in the ICM that modifies
hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:
$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):
$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \qquad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{–}4\%$
at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm
buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$,
suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25\;\text{THz}$ sets the temporal
scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs
the phonon transmission efficiency across the ICM.

<!-- PKG-YM-S225 -->

Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

> Upgrade from PAPER_1318 (Yang-Mills Mass Gap via SCm BCS Phonon) and
> PAPER_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER_1004
> (QGP Vacuum Density), PAPER_1007 (Deconfinement Phase Diagram),
> PAPER_1059 (CGC BK Saturation), PAPER_1064 (Resummation BFKL/Sudakov).

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot \exp\!\left(-\frac{1}{\alpha_s(T) \cdot N_c}\right) \cdot S_{26}^{(3)}$$

where the running coupling evolves as:
$$\alpha_s(T) = \frac{\alpha_{s,0}}{1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c)}, \qquad b_0 = \frac{11 N_c - 2 N_f}{12\pi}$$

Physical mechanism: The SCm phonon field ($\omega_{\text{SCm}} = 1.25\;\text{THz}$) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap $\Delta_{\text{YM}} \approx 1.736\;\text{GeV}$ (PAPER_1318 integer-primitive closure; lattice QCD anchor 1.7 GeV; supersedes 1.736 GeV registry-bug value).

VDS bridge (PAPER_1070): The vacuum density series links the gap to the 26-level hierarchy: $\Delta \propto \rho_{\text{VDS}}^{1/4} \cdot (1 + [\text{SSq}] \cdot n/26)$ where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

QGP transition (PAPER_1004/1007): At $T > T_c \approx 170\;\text{MeV}$, the phonon coupling weakens ($\alpha_s \to 0$) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
$h_\text{UQFF}/h_\text{GR}$	—	$1 - \phi_\text{phonon}$	Strain correction
ϕ_phonon	—	$\sim 10^{-4}$	Phonon phase shift

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
h_UQFF strain	$h_\text{GR}(1 - \phi_\text{phonon})$	Derived
λ_UQFF tidal	Modified by phonon EOS	Novel
Mass gap probability	P_NS/P_BH from ϕ_phonon	Predicted

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification **Sector:** Gravitational Wave / NS Phonon (GW190425 Strain Correction)

§A.2 Lagrangian Density $$\mathcal{L}_\text{GW} = \frac{c^2}{16\pi G}\left(\partial_mu h_{\alpha\beta}\right)^2 + \mathcal{L}_\text{phonon}(\phi_text{phonon}, \Delta)$$

§A.3 Euler-Lagrange Equation of Motion $$\boxed{h_\text{UQFF} = h_\text{GR}\left(1 - \frac{\Delta}{\hbar\omega_\text{SCm}} \cdot S_{26} \cdot \frac{F_{UBi}}{F_U}\right)}$$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\to$ SCm vacuum $\to$ BCS gap $\to$ phonon phase ϕ $\to$ GW strain correction $\to$ GW190425 observable

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS $\phi_text{phonon}$ is proportional to VDS inside the merging neutron stars.

§B.2 DVP GW polarization modes couple to dipole vortex orientation.

§B.3 BSH $h_\text{UQFF}/h_\text{GR}$ bounded in $[1-\phi_text{max}, 1]$ (BSH envelope).

§B.4 Production-Scale Consistency

Metric	Value	Status
$\phi_text{phonon}$	$\sim 10^{-4}$	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_U_Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction

PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1318	Yang-Mills Mass Gap via SCm BCS Phonon Coupling
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1035	Kilonova Buoyancy Light Curve r-Process
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

24 cross-reference(s) identified.

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